Electrical/Control Technical Report

MSE 312

July 21, 2025

Group 14

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Abstract

This project involved designing, building, and testing a circuit and control system to operate a DC motor for a moving mechanical arm. The motor's direction was controlled using a full H-bridge driver, whereas the speed of the arm was adjusted by changing the Pulse Width Modulation (PWM) signals from the Arduino microcontroller. Logic gates handled control signals, and capacitors were added to the circuit to stabilize the power supply and reduce noise. A PID controller was developed in MATLAB/Simulink to manage the angular position and speed of the motor using encoder feedback. This controller used proportional, integral, and derivative components to decrease error and increase accuracy. One of the major challenges faced during testing was tuning the PID values because it required using the trial and error method, which made the testing quite frustrating and time consuming. Another issue that made the tuning of the PID controller challenging was that not all the balls weighed the exact same, which meant that each ball of a different mass had to be tuned to the system or else it would undershoot or overshoot the intended target. Further challenges also came in the form of loose wires, inconsistent power delivery, or overheating of the motor driver. To fix these issues, all connections were rechecked and fixed if needed, the voltage was monitored using an oscilloscope, and a heat sink was installed. Additionally, the testing of the circuit was guided by a series of Integration labs that helped build and verify each part of the project step-by-step. Ultimately, the system was able to move the motor, respond to input changes, and throw the ball, but the response is still slow and prone to error. To improve the performance, further testing and tuning must be done in order to improve the system's speed and accuracy.

1. Introduction

In this project, the focus was on designing and building a circuit that could control the speed and direction of a DC motor. The motor was driven using a full H-bridge driver, which is what enabled the mechanical arm to rotate in both forward and backward directions. The speed of the arm was controlled by changing the Pulse Width Modulation (PWM) signal from the Arduino microcontroller. Logic gates were used to manage control signals, and capacitors were added to the circuit to stabilize the power supply and reduce noise. The circuit was also thoroughly tested to ensure that it worked under different speeds and directions, and that it could handle changes in motor power or behaviour. Furthermore, MATLAB/Simulink were used to simulate the system and analyze how it responded to different inputs. This report provides a detailed explanation of the circuit design, testing procedures, and overall performance based on both the simulation and experimental results.

2. Electrical Circuitry Design

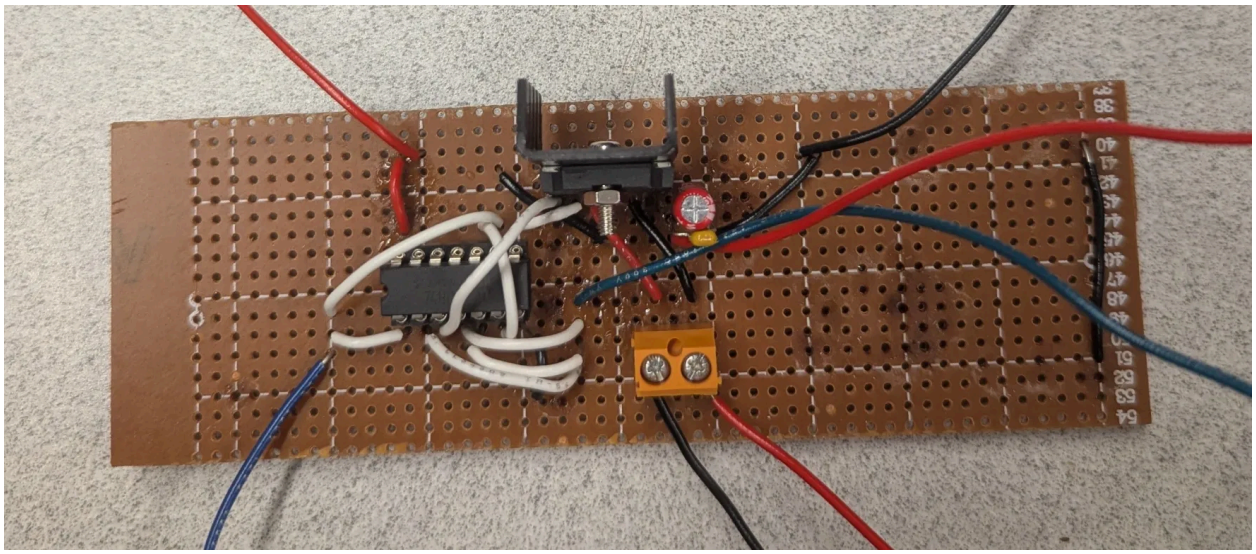


Figure 1. Electronic circuit with heat sink.

Power is supplied to the circuit through the screw terminal block located at the bottom right. The red wire attached to it is connected to motor power and the black wire is connected to motor ground. This power input feeds both the motor driver and capacitor filter network. On the top right, the full-bridge DC motor driver IC is connected to a THM6073 heat sink, which is used to dissipate all the heat generated by the current going through the circuit. It receives 12V of power and ground through the red and black wires, respectively.

Control logic for the motor driver is provided by a Quad 2-input NAND gate located on the middle left of the board. The NAND gate processes logic inputs and sends output signals to the motor driver through white connection wires. The blue wire on the left is connected to Pin 45 and the blue wire on the right is connected to Pin 44. Both these wires supply the logic inputs to the NAND gate from the Arduino microcontroller. In addition, to supply voltage to the circuit, the red wire on the top left is connected to the power terminal of the Arduino.

To ensure voltage stability, a red 10µF capacitor is placed next to the motor driver. This capacitor filters out low-frequency noise and protects the driver from voltage spikes. Alongside it, a yellow 100nF radial ceramic capacitor is used to filter high-frequency noise and respond to fast, small voltage spikes that the larger capacitor might miss. Together, these capacitors form a complementary filtering network that stabilizes the power supply and reduces noise, thereby increasing the circuit's performance.

3. Control System Design

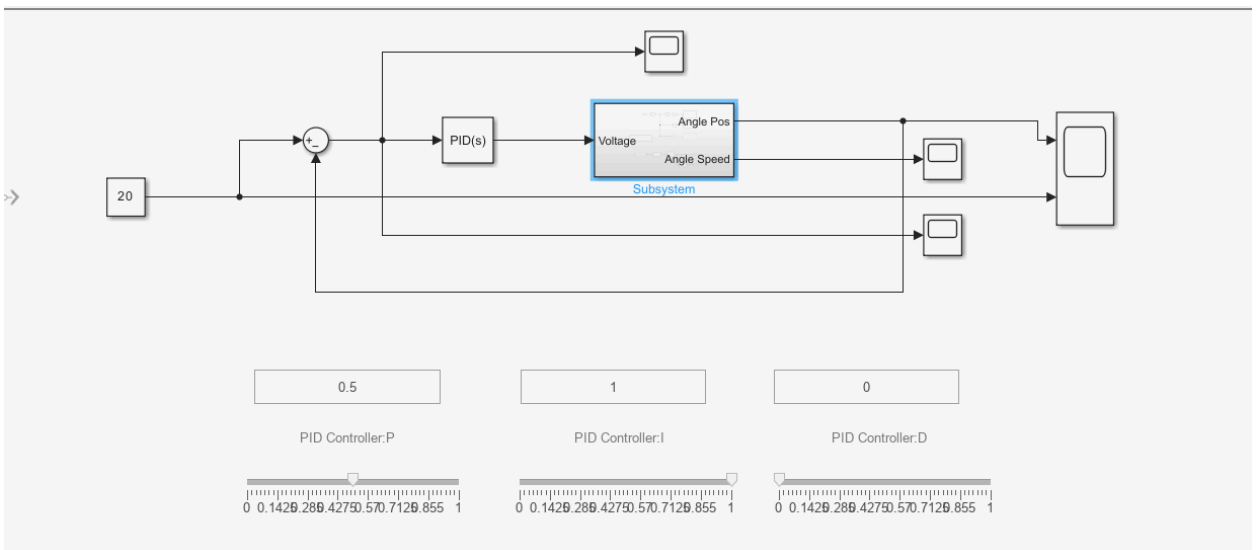


Figure 2. Simulink model of PID based control system.

The current design for the control system is a PID based system. PID stands for proportional, integral, and derivative. The combination of these systems allow a system to reach a steady value, and stay there if conditions change. The proportional component reacts to the error in the element being tracked. The integral element reacts to the error which accumulates over time in the element being tracked. The derivative element predicts future error in the element being tracked. This is done by tracking the rate of change of the error.

The input for the system is voltage and the output is motion of the motor. The output is the angular position and angular speed of the motor. A constant is used as a temporary voltage for the motor. This is then connected to a PID block. The characteristics of the PID are then controlled by the sliders at the bottom. Using the voltage, and the required angle of the motor, the system will adjust the motor voltage to properly place the angle of the motor at the pre-set value.

For testing purposes, multiple scopes were used. This was to see how the voltage changed over time, how the angular position changed over time, how the error changed over time, and how the angular speed changed over time. With all the information provided, the behaviour of the PID system can be calibrated.

4. Electric Circuit Insights & Challenges

One of the main challenges encountered in the electrical circuit part of the project was getting the arm to reliably move forwards and backwards. This feature is reliant on the functionality of the full-bridge motor driver in tandem with the logic input and input gate components and was resolved through debugging of the logic input. The circuit would also stop working without any warning even when no changes had been made to the code or wiring. This all suggests that the system was unstable and sensitive to small changes. It was assumed this was likely due to loose wires, inconsistent power delivery, or overheating of the motor driver. To fix this issue, firstly, the team checked all the wires in the circuit to confirm they were fixed solidly in place, and if they were loose, that issue was rectified. Then, to handle the matter of the inconsistent power supply, the team ensured a steady supply of 12V was being delivered to the circuit by monitoring it with an oscilloscope and double-checking that all the ground connections were firmly in place. The issue turned out to be the terminal block connection, this was mitigated by omitting it entirely and directly connecting the circuit to the power supply. A key improvement of the circuit was the addition of the heat sink onto the motor driver IC. This allowed for better thermal regulation and ensured that the driver could reliably supply the required peak current of 2.77A without overheating. Although the PWM signal graphs were unable to be included from the initial testing due to file saving issues, the relationship between PWM duty cycle and motor performance was evident through constant trial runs and testing. The performance of the circuit clearly showed that higher duty cycles produced faster motor rotation, while the lower duty cycles caused the motor to slow down, confirming that the motor driver and control logic successfully worked together to meet the design requirements.

5. Control System Insights & Challenges

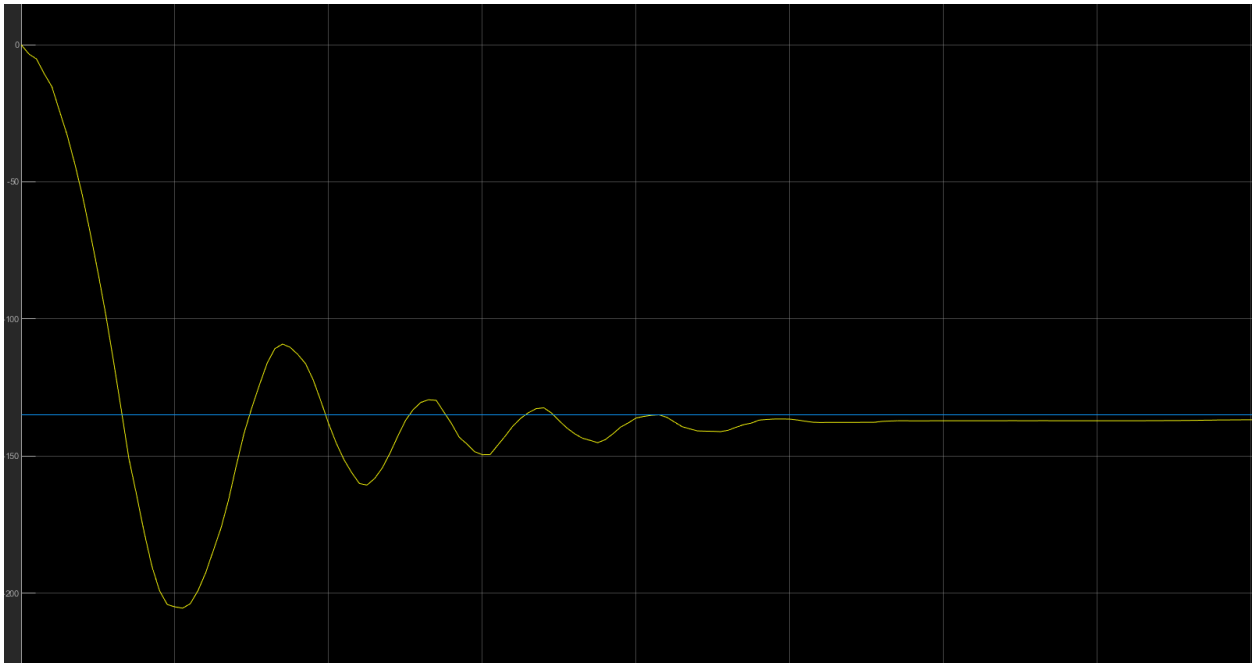


Figure 3. PID based control system response.

Some insights that are made is that the response time is critical. If the reaction is too slow or too fast, the system will not throw the ball into the cup. As a result, minimizing delay is key for the system. Another insight is due to the mass of the ball. Not all balls will weigh the same, so tuning the system to the ball is critical. Using incorrect programming will result in the ball overshooting or undershooting the target.

Our control system was implemented using a PID controller block in Simulink, without incorporating any explicit transfer functions or using model-based tuning techniques as it was determined that manually using the trial and error approach, and tuning parameters based on the output behavior of the system was a viable method. The biggest challenge faced was to properly tune the values of the PID. It is done by a trial and error method, meaning that values are set and tried. If they do not work, then another set of values is tested. This makes the testing quite frustrating, limiting, and challenging. Better calculation or computer simulation of the given parameters would make the tuning of the system much faster. Initially the idea of tuning/adding D (derivative) of the PID controller was the method to improve settling time and reduce overshoot of the system, however we faced quite a few challenges when it came to this parameter. The addition of derivative gain caused excessive motor vibration and instability rendering the motor incapable of following simple movement commands. This was likely due to the increased noise effects stemming from the PWM switching. In addition, we tried to introduce

a lead compensator to the system as a solution to the same effect of the adding derivative gain, but it also produced the same type of vibrations.

Due to the issues regarding those previous potential solutions, we decided to focus on tuning only the proportional and integral gains in order to balance the responsiveness and stability of the system. After extensive tuning, we identified an optimal combination of **P = 0.65** and **I = 0.35**. This combination proved to provide the best overall performance, outputting a moderate overshoot at approx. 30 % and roughly 3-4 oscillations before settling time. The steady state error of the system was minimized by the tuning of I while ensuring that the response did not end up being too slow. While tuning P, we wanted our system to have a fast and responsive reaction without pushing the system to be unstable. The value of 0.65 provided the best balance of achieving the desired response output while minimizing the overshoot as much as possible. If more time was allowed, the next step would be to discover a solution to the derivative gain and vibrations in order to further improve the overshoot and settling time as we have a value of zero for the derivative gain in our PID system. The PID values found in the Simulink volume attached do not correctly represent the values ultimately chosen.

6. Testing Procedures

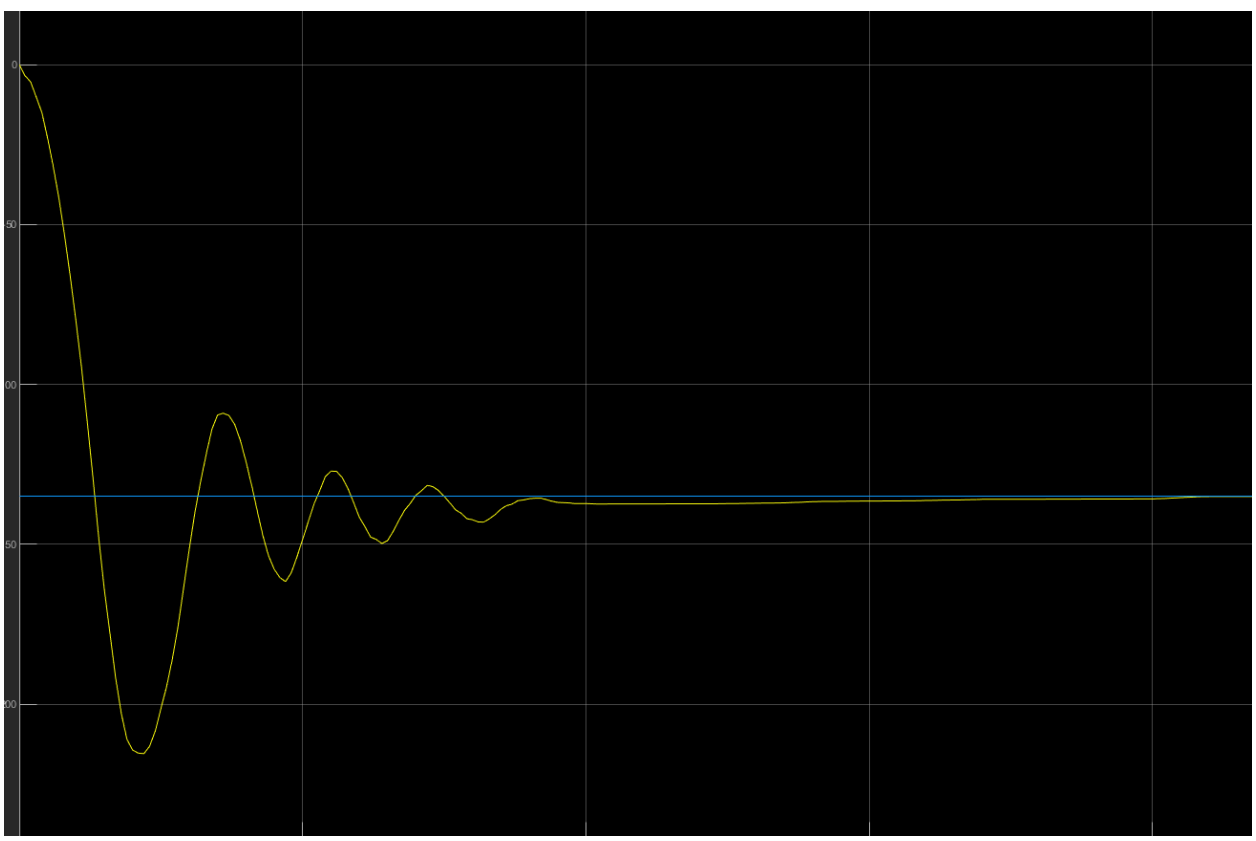


Figure 4. Control system test response.

To test the motor drive and control system, the procedures laid out in the Integration Labs were closely followed. Each successive lab was used to build and check different parts of the final system. In Lab 1, the team learned how to connect MATLAB/Simulink to the Arduino microcontroller and run basic programs to read and write signals. This lab helped confirm that communication between MATLAB/Simulink and Arduino worked properly, verifying a correct setup of hardware and software. This meant that the system could interact with sensors and actuators, which was critical to establish before adding in motor control.

In Lab 2, there was a focus on reading data from a rotary encoder to measure how much the motor turned. This was important because being able to measure the motor shaft position while it was in operation verified that feedback had been added successfully to the control system, which was needed in order to accurately track the position of the motor and help with PID tuning later down the line.

In Lab 3, PWM signals and a H-bridge circuit were used to control the motor's speed and direction. This was necessary in order to vary the motor speed and direction. This lab also verified that the duty cycles could be tuned, that the correct PWM signals were being generated, and that the circuit could drive the motor within its voltage and current limits. All these tests confirmed that the motor could run safely and reliably and respond to control inputs correctly.

Finally, Lab 4 focused on designing, implementing, and tuning a PID controller using encoder feedback to build a working closed-loop control system. It also touched upon trapezoidal trajectory planning and step response testing. In essence, this lab tested how the system responded to different input signals (step, sine, pulse) and different gain settings and adjusted the controller settings to improve accuracy and stability. Altogether, the Integration labs provided a step-by-step guide to test the control system and ensure it met the project goals.

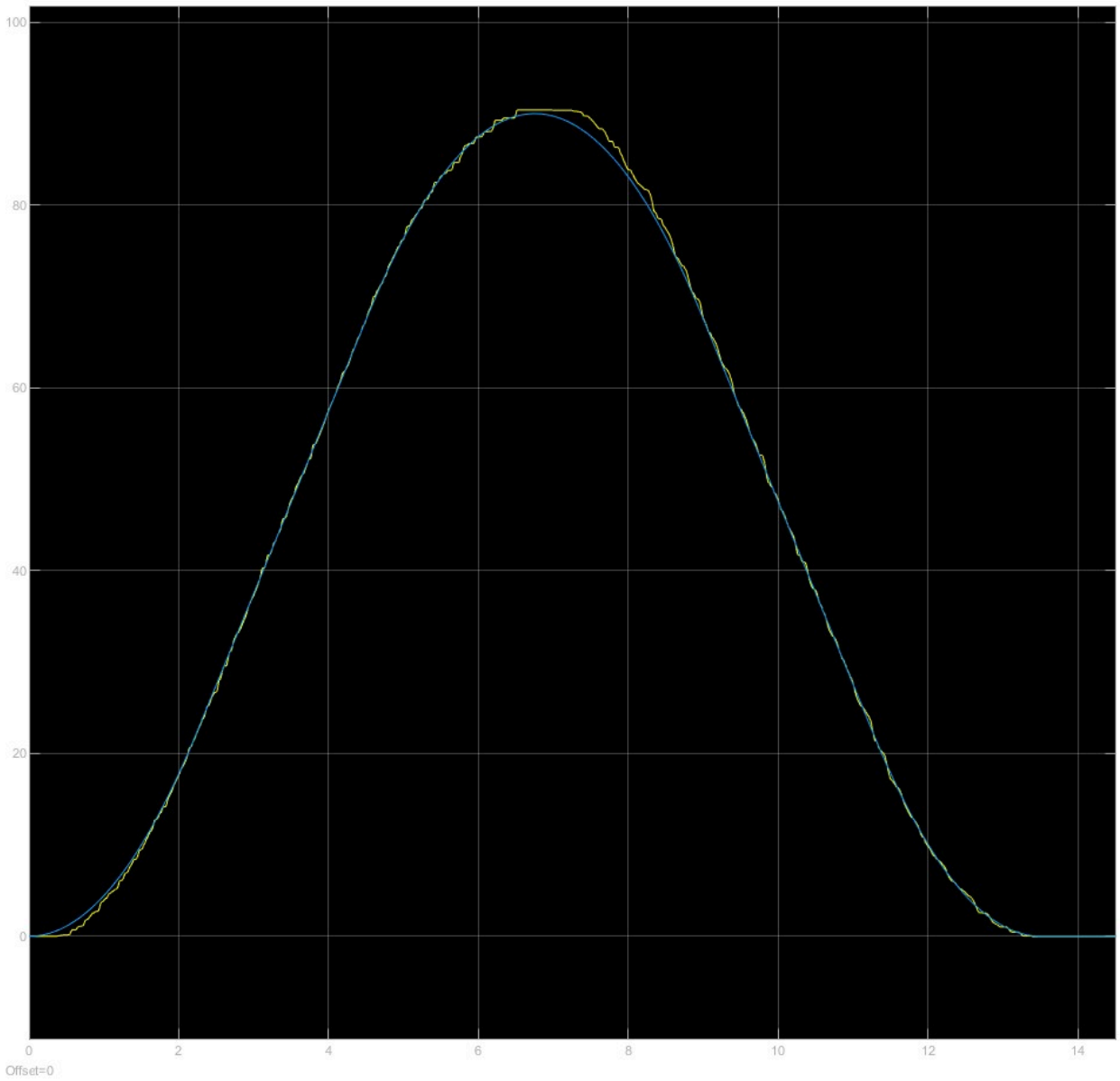
7. Conclusion

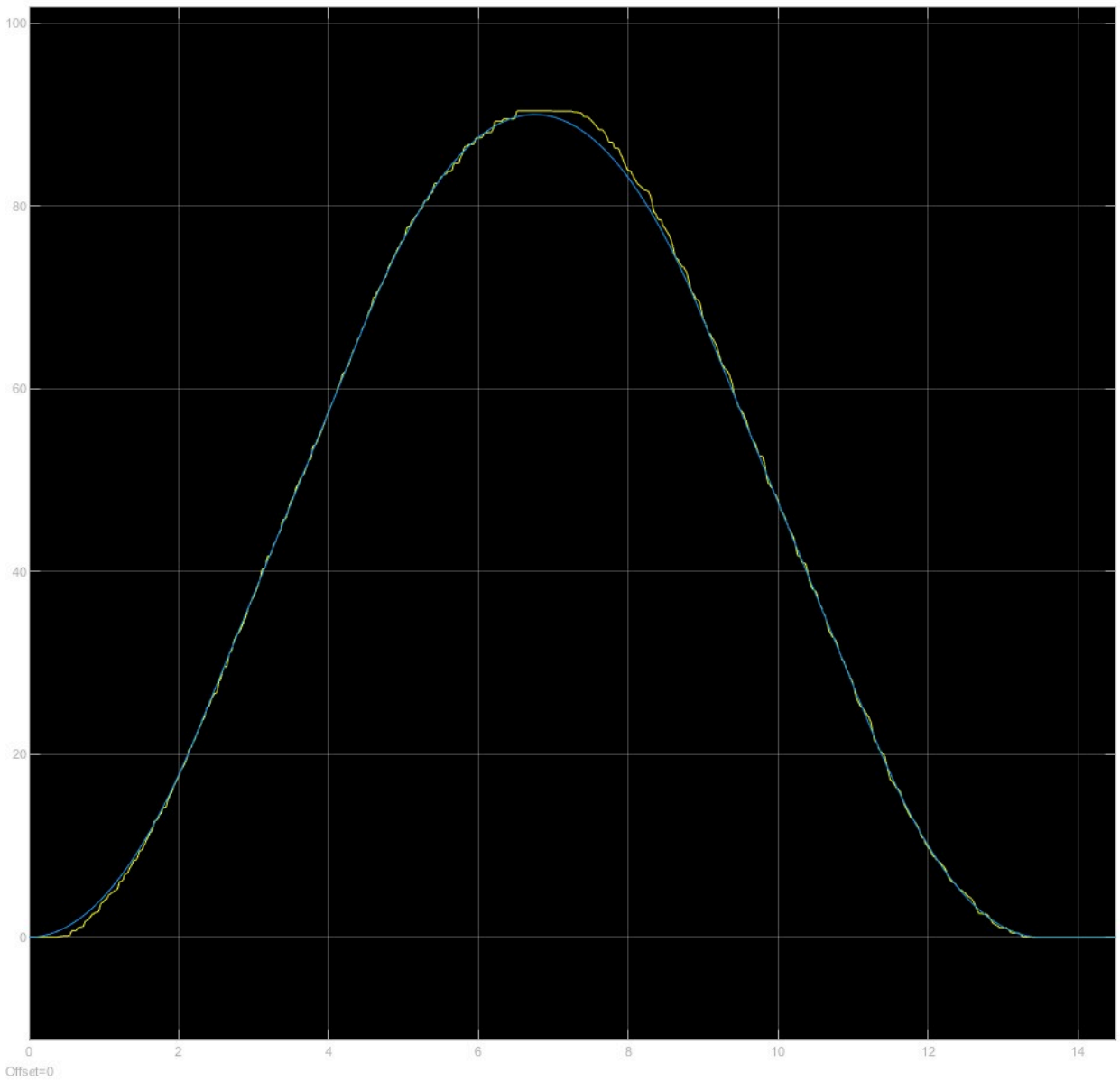
This report spoke about the electronic circuit being used to throw the ball into the target as well as the control system being used to ensure proper accuracy in throwing the ball. The system uses capacitors, logic gates, and transistors to ensure proper power supply and switching of voltage to ensure full control of the motor. The control system utilizes PID to control the target angle and speed of the motor to throw the ball. The testing done so far has rendered the electronic circuit to be working well and responsive to controlling the motor direction and showcasing the position and speed of the motor. The PID testing has shown that the motor does respond and adjust itself to the proper position but the response is still slow and with error still.

Repeated testing will be needed to perfect the PID system to perform as expected. Testing will consistently be done to ensure the circuit works as expected.

8. Appendix

Additional PID Test Results:

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